**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**HIGHER SECONDARY SCHOOL CERTIFICATE**

**CLASS XII**

**ANNUAL EXAMINATIONS (THEORY) 2025**

**Computer Science Paper II**

**Time: 1 hour 30 minutes  Marks: 35**

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**INSTRUCTIONS**

**Please read the following instructions carefully**.

1. Check your name and school information. Sign if it is accurate.

| **I agree that this is my name and school.**<br>**Candidate's Signature** |
|---|

**RUBRIC**

2. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 3 Marks)

Write any THREE differences between Command Line Interface (CLI) and Graphical User Interface (GUI).

| S.No. | CLI | GUI |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

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Q.2. (Total 3 Marks)

Describe the responsibilities of any THREE of the following personnel in System Development Life Cycle (SDLC).

- Management team
- System analyst
- Project manager
- Programmer
- Software tester

Q.3. (Total 5 Marks)

i. Write a program to display the following output. (2 Marks)

```
This
is
a
test        program

He said, "How can a clam cram in a clean cream can?"
```

ii. Write a program to find the largest number among 1, 2 and 5. (3 Marks)

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Q.4. (Total 3 Marks)

Write any THREE differences between while and do while loop.

| S.No. | While Loop | Do While Loop |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

Q.5. (Total 4 Marks)

Write a C++ program to illustrate single inheritance. The program should include the following:

- A base class named Company with a variable named CompanyName that stores the string "Softech Innovators Ltd."
- A derived class named Employee that inherits from the Company. It should also store the string "Ahmed Raza" in the variable EmployeeName, "Engineering" in the variable Department, and "Software Engineer" in the variable JobTitle.
- Create an object of the Employee class named empObj to display the company name, employee name, department, and job title.
- The output should be the same as shown below:

```
Company Name: Softech Innovators Ltd.
Employee Name: Ahmed Raza
Department: Engineering
Job Title: Software Engineer
```

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Q.6. (Total 3 Marks)

Write a C++ program that

- creates a text file named as cityname.txt
- writes following line in the file.
  "Karachi is the capital of Sindh"
- closes the file.

Q.7. (Total 7 Marks)

**EITHER**

a. Write a C++ program for calculating and printing the surface area and volume of a sphere using a formula.

Given is the formula to calculate the surface area and volume of a sphere.

- Surface Area = $4\pi r^2$
- Volume of Sphere = $4/3\pi r^3$

(**Note:** The value of $\pi = 3.142$ is declared as symbolic constant.)

**OR**

b. Consider a simple geometric calculator that calculates the area of different shapes: circle, square and rectangle.

Write a C++ program for this calculator using switch statement that

- inputs a character representing the shape ('C' for circle, 'S' for square, 'R' for rectangle).
- uses the given formula to calculate the area of circle, square and rectangle.
  - Circle Area = $\pi r^2$
  - Square Area = side * side
  - Rectangle Area = length * width
- inputs the necessary dimensions as float data type for each shape.
- outputs the calculated area of the shape.

(**Note:** The value of $\pi = 3.142$ is declared as symbolic constant.)

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Q.8. (Total 7 Marks)

**EITHER**

a. Write a program using a user-defined function named 'fibonacci' that would generate the fibonacci series in which each term is the sum of the previous two terms till the end of the selected range and invoke it in the main function.

- The formula of fibonacci series is:

  $F(n) = F(n-1) + F(n-2)$

- With the initial values: $F(0) = 0$, $F(1) = 1$

**OR**

b. Write a C++ program that reverses an inputted string of 10 alphabets and prints it on the screen.

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END OF PAPER

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